

Custom MapReduce with Hadoop Streaming for Online Food Delivery Analytics

Project Objective

Analyze online food delivery transactions using Hadoop Streaming to:

- Find city-wise revenue
- Identify top-selling food items
- Analyze monthly revenue trends
- Determine average order values
- Visualize business insights using Python

Part 1: Generate Food Delivery Dataset

nano generate_food_orders.py

```
import random
import datetime
```

```
foods = [
    "Pizza", "Burger", "Pasta", "Sandwich",
    "Biryani", "Noodles", "Fries", "Momos"
]
```

```
cities = [
    "Mumbai", "Delhi", "Bangalore",
    "Chennai", "Hyderabad", "Pune",
    "Ahmedabad", "Kolkata"
]
```

```

customers = [
    f'CUST{str(i).zfill(5)}'
    for i in range(1, 2001)
]

start_date = datetime.date(2025, 1, 1)
end_date = datetime.date(2025, 12, 31)

days = (end_date - start_date).days

print("Date,FoodItem,Price,City,CustomerID,Quantity")

for _ in range(60000):

    date = start_date + datetime.timedelta(
        days=random.randint(0, days)
    )

    food = random.choice(foods)

    price = round(
        random.uniform(80, 1200),
        2
    )

    city = random.choice(cities)

    customer = random.choice(customers)

    quantity = random.randint(1, 6)

    print(
        f'{date},{food},{price},{city},{customer},{quantity}'
    )

python3 generate_food_orders.py > food_orders.csv

ls -lh food_orders.csv
wc -l food_orders.csv
head -5 food_orders.csv

```

Part 2: Mapper - City Revenue

nano mapper_city_revenue.py

```
#!/usr/bin/env python3
```

```
import sys
```

```
next(sys.stdin)
```

```
for line in sys.stdin:
```

```
    parts = line.strip().split(",")
```

```
    if len(parts) != 6:  
        continue
```

```
    city = parts[3]
```

```
    try:  
        price = float(parts[2])  
        qty = int(parts[5])
```

```
        revenue = price * qty
```

```
        print(f'{city}\t{revenue}')
```

```
    except:  
        continue
```

Part 3: Reducer - City Revenue

nano reducer_city_revenue.py

```
#!/usr/bin/env python3
```

```
import sys
```

```
current_city = None
```

```
total_revenue = 0
```

```
orders = 0
```

```
for line in sys.stdin:
```

```

city, revenue = line.strip().split("\t")

revenue = float(revenue)

if city == current_city:

total_revenue += revenue
    orders += 1

else:

    if current_city:

print(
    f"{current_city}\t{total_revenue:.2f}\t{orders}"
)

current_city = city
total_revenue = revenue
orders = 1

if current_city:

print(
    f"{current_city}\t{total_revenue:.2f}\t{orders}"
)

```

Part 4: Local Testing

```

head -20 food_orders.csv \
| python3 mapper_city_revenue.py \
| sort \
| python3 reducer_city_revenue.py

```

Part 5: Upload To HDFS

```
hdfsdfs -mkdir -p /user/$USER/food/input
```

```
hdfsdfs -put food_orders.csv \
/user/$USER/food/input/
```

```
hdfsdfs -ls /user/$USER/food/input
```

Part 6: Run Hadoop Streaming

```
hadoop jar \  
$HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \  
-input /user/$USER/food/input/food_orders.csv \  
-output /user/$USER/food/output \  
-mapper "python3 mapper_city_revenue.py" \  
-reducer "python3 reducer_city_revenue.py" \  
-file mapper_city_revenue.py \  
-file reducer_city_revenue.py
```

Part 7: Download Results

```
hdfsdfs -get \  
/user/$USER/food/output/part-00000 \  
city_revenue.tx
```

```
cat city_revenue.txt
```

```
sort -t'$\t' -k2 -rn city_revenue.txt | head -5
```

Part 8: Monthly Revenue Analysis

```
nano mapper_monthly_revenue.py
```

```
#!/usr/bin/env python3
```

```
import sys
```

```
next(sys.stdin)
```

```
for line in sys.stdin:
```

```
    parts = line.strip().split(",")
```

```
    date = parts[0][:7]
```

```
    revenue = (  
        float(parts[2])  
        * int(parts[5])  
    )
```

```
    print(f'{date}\t{revenue}')
```

```
#!/usr/bin/env python3

import sys

current_month = None
total = 0

for line in sys.stdin:

    month, revenue = line.strip().split("\t")

    revenue = float(revenue)

    if month == current_month:

        total += revenue

    else:

        if current_month:

            print(
                f'{current_month}\t{total:.2f}'
            )

        current_month = month
        total = revenue

    if current_month:

        print(
            f'{current_month}\t{total:.2f}'
        )
```

Part 9: Food Item Quantity Analysis

```
#!/usr/bin/env python3

import sys

next(sys.stdin)

for line in sys.stdin:
```

```
parts = line.strip().split(",")
```

```
item = parts[1]
```

```
qty = int(parts[5])
```

```
print(f'{item}\t{qty}')
```

Screenshots

```
sky@MS97377: ~  
sky@MS97377:~$ cat > generate_sales.py << 'EOF'  
>  
> import random  
import d> import datetime  
>  
>  
>  
>  
> cat > generate_sales.py << 'EOF'  
import random  
import datetime  
>  
>  
>  
>  
^C  
sky@MS97377:~$ cat > generate_sales.py << 'EOF'  
>  
> import random  
> import datetime  
>  
>  
>  
> products = ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones', 'Webcam', 'Printer', 'Scanner']  
ies = ['>  
> cities = ['Mumbai', 'Delhi', 'Bangalore', 'Chennai', 'Hyderabad', 'Pune', 'Kolkata', 'Ahmedabad']  
> customers = [f'CUST{str(i).zfill(5)}' for i in range(1, 1001)]  
> start_date = datetime.date(2024, 1, 1)  
> end_date = datetime.date(2024, 12, 31)  
> date_range = (end_date - start_date).days  
> for i in range(50000):  
sky@MS97377:~$ cat > generate_sales.py << 'EOF'  
>  
> import random  
> import datetime  
>  
>  
>  
> products = ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones', 'Webcam', 'Printer', 'Scanner']  
ies = ['>  
> cities = ['Mumbai', 'Delhi', 'Bangalore', 'Chennai', 'Hyderabad', 'Pune', 'Kolkata', 'Ahmedabad']  
> customers = [f'CUST{str(i).zfill(5)}' for i in range(1, 1001)]  
> start_date = datetime.date(2024, 1, 1)  
> end_date = datetime.date(2024, 12, 31)  
> date_range = (end_date - start_date).days  
> for i in range(50000):  
    date = start_date + datetime.timedelta(days=random.randint(0, date_range))  
    product = random.choice(products)  
price =>    price = round(random.uniform(100, 5000), 2)  
= random>    city = random.choice(cities)  
stomer =>    customer = random.choice(customers)  
>    quantity = random.randint(1, 5)  
>    print(f'{date},{product},{price},{city},{customer},{quantity}')> EOF  
sky@MS97377:~$ python3 generate_sales.py > sales_data.csv  
sky@MS97377:~$  
sky@MS97377:~$ ls -lh sales_data.csv  
-rw-r--r-- 1 sky sky 2.3M Jul 25 16:13 sales_data.csv  
sky@MS97377:~$ wc -l sales_data.csv  
50000 sales_data.csv  
sky@MS97377:~$
```

```

sky@MS97377: ~
50000 sales_data.csv
sky@MS97377:~$
sky@MS97377:~$ head -5 sales_data.csv
2024-01-03,Scanner,4686.88,Chennai,CUST00786,5
2024-08-03,Printer,1146.62,Mumbai,CUST00269,5
2024-09-17,Laptop,3177.58,Mumbai,CUST00666,2
2024-11-18,Headphones,2837.51,Mumbai,CUST00692,4
2024-02-16,Headphones,1555.84,Chennai,CUST00560,1
sky@MS97377:~$ cat sales_data.csv | python3 mapper_city_sales.py
python3: can't open file '/home/sky/mapper_city_sales.py': [Errno 2] No such file or directory
sky@MS97377:~$ cat > mapper_city_sales.py << 'EOF'
> #!/usr/bin/env python3
> sys
>
> for
> import sys
>
> for line in sys.stdin:
>     # Skip empty lines
>     line = line.strip()
>     if not line:
>         continue
>
>     # Parse CSV: date,product,price,city,customer,quantity
> parts = line.split(',')
>
> if len(parts) != 6:
>     continue
>
>     try:
>         date = parts[0]
>         product = parts[1]
>         price = float(parts[2])
>         city = parts[3]
>         customer = parts[4]
>         quantity = int(parts[5])
>
>         # Calculate total sales amount
>         total = price * quantity
>
>         # Emit city and total sales
>         print(f"{city}\t{total}")
>
> except ValueError:
>     continue
sky@MS97377:~$
>
>     try:
>         date = parts[0]
>         product = parts[1]
>         price = float(parts[2])
>         city = parts[3]
>         customer = parts[4]
>         quantity = int(parts[5])
>
>         # Calculate total sales amount
>         total = price * quantity
>
>         # Emit city and total sales
>         print(f"{city}\t{total}")
>
> except ValueError:
>     continue
> EOF
sky@MS97377:~$ chmod +x mapper_city_sales.py
sky@MS97377:~$ ls -l mapper_city_sales.py
-rwxr-xr-x 1 sky sky 633 Jul 25 16:19 mapper_city_sales.py
sky@MS97377:~$ head -10 sales_data.csv | python3 mapper_city_sales.py
Chennai 23434.4
Mumbai 5733.099999999999
Mumbai 6355.16
Mumbai 11350.04
Chennai 1555.84
Ahmedabad 8078.44
Bangalore 7772.06
Pune 11937.27
Delhi 6143.84
Ahmedabad 1779.8500000000001
sky@MS97377:~$ cat > reducer_city_sales.py << 'EOF'
> /usr/bin/#!/usr/bin/env python3
>
> import sys
>
> current_city = None
> total_sales = 0
> count = 0
>
> for line in sys.stdin:
>     line = line.strip()

```



```

sky@MS97377:~$ chmod +x mapper_city_sales.py
sky@MS97377:~$ ls -l mapper_city_sales.py
-rwxr-xr-x 1 sky sky 633 Jul 25 16:19 mapper_city_sales.py
sky@MS97377:~$ head -10 sales_data.csv | python3 mapper_city_sales.py
Chennai 23434.4
Mumbai 5723.099999999999
Mumbai 6355.16
Mumbai 11350.04
Chennai 1555.84
Ahmedabad 8078.44
Bangalore 7772.06
Pune 11937.27
Delhi 6143.84
Ahmedabad 1779.8500000000001
sky@MS97377:~$ cat > reducer_city_sales.py << 'EOF'
#!/usr/bin/env python3
>
> import sys
>
> current_city = None
> total_sales = 0
> count = 0
>
> for line in sys.stdin:
>     line = line.strip()
>
>     if not line:
>         continue
city, am>
>     city, amount = line.split('\t')
>     amount = float(amount)
>
>     if c>
>         if city == current_city:
>             total_sales += amount
>             count += 1
>         else:
>             if current_city is not None:
>                 print(f"{current_city}\t{total_sales:.2f}\t{count}")
>             cur>
>             current_city = city
>         total>
>         total_sales = amount
>         count = 1
>
>
> print las> # Print last city
sky@MS97377:~$
>         continue
city, am>
>     city, amount = line.split('\t')
>     amount = float(amount)
>
>     if c>
>         if city == current_city:
>             total_sales += amount
>             count += 1
>         else:
>             if current_city is not None:
>                 print(f"{current_city}\t{total_sales:.2f}\t{count}")
>             cur>
>             current_city = city
>         total>
>         total_sales = amount
>         count = 1
>
>
> print las> # Print last city
f curren> if current_city is not None:
>     print(f"{current_city}\t{total_sales:.2f}\t{count}")
>
>
EOF> EOF
sky@MS97377:~$ chmod +x reducer_city_sales.py
sky@MS97377:~$ ls -l reducer_city_sales.py
-rwxr-xr-x 1 sky sky 602 Jul 25 16:22 reducer_city_sales.py
sky@MS97377:~$ head -20 sales_data.csv | python3 mapper_city_sales.py | sort | python3 reducer_city_sales.py
Ahmedabad 20552.81 4
Bangalore 8525.14 2
Chennai 38118.48 5
Delhi 9026.49 2
Hyderabad 15721.20 2
Kolkata 2631.71 1
Mumbai 23438.30 3
Pune 11937.27 1
sky@MS97377:~$ head -20 sales_data.csv | python3 mapper_city_sales.py | sort | python3 reducer_city_sales.py
Ahmedabad 20552.81 4
Bangalore 8525.14 2
Chennai 38118.48 5
Delhi 9026.49 2
Hyderabad 15721.20 2
Kolkata 2631.71 1
Mumbai 23438.30 3
Pune 11937.27 1
sky@MS97377:~$ jps
779 jps

```

```

sky@MS97377: ~
/*****
SHUTDOWN MSG: Shutting down NameNode at MS97377/127.0.1.1
*****/
sky@MS97377:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [MS97377]
sky@MS97377:~$ jps
2401 NameNode
2852 Jps
2710 SecondaryNameNode
2518 DataNode
sky@MS97377:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
sky@MS97377:~$ jps
3424 Jps
2401 NameNode
3059 NodeManager
2949 ResourceManager
2710 SecondaryNameNode
2518 DataNode
sky@MS97377:~$ hdfs dfs -mkdir -p /user/$USER/sales/input
sky@MS97377:~$ hdfs dfs -mkdir -p /user/$USER/sales/output
sky@MS97377:~$ hdfs dfs -put sales_data.csv /user/$USER/sales/input/
sky@MS97377:~$ hdfs dfs -ls /user/$USER/sales/input/
Found 1 items
-rw-r--r-- 1 sky supergroup 2335554 2026-07-25 16:37 /user/sky/sales/input/sales_data.csv
sky@MS97377:~$ jps
2401 NameNode
3059 NodeManager
2949 ResourceManager
2710 SecondaryNameNode
2518 DataNode
3623 Jps
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
put /use> -input /user/$USER/sales/input/sales_data.csv \
ut /user> -output /user/$USER/sales/output \
> -mapper "python3 mapper_city_sales.py" \
> -reducer "python3 reducer_city_sales.py" \
file map> -file mapper_city_sales.py \
> -file reducer_city_sales.py
2026-07-25 16:38:36,255 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_city_sales.py, reducer_city_sales.py, /tmp/hadoop-unjar7806270873359042940/] [] /tmp/streamjob1274348300271297012.jar tmpDir=null
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
put /use> -input /user/$USER/sales/input/sales_data.csv \
ut /user> -output /user/$USER/sales/output \
> -mapper "python3 mapper_city_sales.py" \
> -reducer "python3 reducer_city_sales.py" \
file map> -file mapper_city_sales.py \
> -file reducer_city_sales.py
2026-07-25 16:38:36,255 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_city_sales.py, reducer_city_sales.py, /tmp/hadoop-unjar7806270873359042940/] [] /tmp/streamjob1274348300271297012.jar tmpDir=null
2026-07-25 16:38:37,962 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:38:38,304 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:38:38,510 ERROR streaming.StreamJob: Error Launching job : Output directory hdfs://localhost:9000/user/sky/sales/output already exists
Streaming Command Failed!
sky@MS97377:~$ hdfs dfs -rm -r /user/$USER/sales/output
Deleted /user/sky/sales/output
sky@MS97377:~$ hdfs dfs -ls /user/$USER/sales
Found 1 items
drwxr-xr-x 1 sky supergroup 0 2026-07-25 16:37 /user/sky/sales/input
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
> -input /user/$USER/sales/input/sales_data.csv \
> -output /user/$USER/sales/output \
> -mapper "python3 mapper_city_sales.py" \
> -reducer "python3 reducer_city_sales.py" \
> -file mapper_city_sales.py \
file redu> -file reducer_city_sales.py
2026-07-25 16:40:17,033 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_city_sales.py, reducer_city_sales.py, /tmp/hadoop-unjar716032600467264502/] [] /tmp/streamjob2661018985131943659.jar tmpDir=null
2026-07-25 16:40:19,538 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:40:19,933 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:40:20,348 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1784977574676_0001
2026-07-25 16:40:21,649 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 16:40:22,234 INFO mapreduce.JobSubmitter: number of splits:2
2026-07-25 16:40:22,988 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1784977574676_0001
2026-07-25 16:40:22,989 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-25 16:40:23,336 INFO conf.Configuration: resource-types.xml not found
2026-07-25 16:40:23,337 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-25 16:40:23,829 INFO impl.YarnClientImpl: Submitted application application_1784977574676_0001
2026-07-25 16:40:23,943 INFO mapreduce.Job: The url to track the job: http://MS97377.localdomain:8088/proxy/application_1784977574676_0001/
2026-07-25 16:40:23,949 INFO mapreduce.Job: Running job: job_1784977574676_0001
2026-07-25 16:40:39,291 INFO mapreduce.Job: Job job_1784977574676_0001 running in uber mode : false
2026-07-25 16:40:39,382 INFO mapreduce.Job: map 0% reduce 0%
2026-07-25 16:40:54,942 INFO mapreduce.Job: map 58% reduce 0%

```

```

sky@MS97377: ~
File Output Format Counters
Bytes Written=200
2026-07-25 16:41:05,464 INFO streaming.StreamJob: Output directory: /user/sky/sales/output
sky@MS97377:~$ hdfs dfs -ls /user/$USER/sales/output
Found 2 items
-rw-r--r-- 1 sky supergroup 0 2026-07-25 16:41 /user/sky/sales/output/ SUCCESS
-rw-r--r-- 1 sky supergroup 200 2026-07-25 16:41 /user/sky/sales/output/part-00000
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/output/part-00000
Ahmedabad 48144482.44 6232
Bangalore 48713073.14 6300
Chennai 48259632.40 6231
Delhi 47550288.23 6188
Hyderabad 47306662.01 6243
Kolkata 47815272.70 6256
Mumbai 47276682.25 6174
Pune 48973969.07 6376
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/output/part-00000 | head
Ahmedabad 48144482.44 6232
Bangalore 48713073.14 6300
Chennai 48259632.40 6231
Delhi 47550288.23 6188
Hyderabad 47306662.01 6243
Kolkata 47815272.70 6256
Mumbai 47276682.25 6174
Pune 48973969.07 6376
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/output/part-00000 | sort -t'$'\t' -k2 -rn | head -5
Pune 48973969.07 6376
Bangalore 48713073.14 6300
Chennai 48259632.40 6231
Ahmedabad 48144482.44 6232
Kolkata 47815272.70 6256
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/output/part-00000 | awk -F'$'\t' '{printf "%s\t%.2f\n", $1, $2/$3}' | sort -t'$'\t' -k2 -rn
Chennai 7745.09
Bangalore 7732.23
Ahmedabad 7725.37
Delhi 7684.27
Pune 7680.99
Mumbai 7657.38
Kolkata 7643.11
Hyderabad 7577.55
sky@MS97377:~$ cat > mapper_monthly_sales.py << 'EOF'
#!/usr/bin/env python3
>
> import sys
>
sky@MS97377:~$ cat > mapper_monthly_sales.py << 'EOF'
#!/usr/bin/env python3
>
> import sys
>
> for line in sys.stdin:
>     line = line.strip()
>
>     if not line:
>         continue
>
>     parts = line.split(',')
>
format:>     if len(parts) != 6:
>         continue
>
>     try:
>         date = parts[0] # Format: YYYY-MM-DD
>         month = date[:7] # Extract YYYY-MM
>         price = float(parts[2])
>         quantity = int(parts[5])
error:>
>         continue
EOF>
>
>         total = price * quantity
>
>         print(f"{month}\t{total}")
>
>     except ValueError:
>         continue
>
> EOF
sky@MS97377:~$ chmod +x mapper_monthly_sales.py
sky@MS97377:~$ cat > reducer_monthly_sales.py << 'EOF'
#!/usr/bin/env python3
>
> import sys
>
> current_month = None
> total = 0
>
> for line in sys.stdin:
>     line = line.strip()

```

```

sky@MS97377: ~
> continue
> EOF
sky@MS97377:~$ chmod +x mapper_monthly_sales.py
sky@MS97377:~$ cat > reducer_monthly_sales.py << 'EOF'
#!/usr/bin/env python3

import sys

current_month = None
total = 0

for line in sys.stdin:
    line = line.strip()

    if not line:
        continue

    month, amount = line.split('\t')
    amount = float(amount)

    if month == current_month:
        total += amount
    else:
        if current_month is not None:
            print(f'{current_month}\t{total:.2f}')

        current_month = month
        total = amount

if current_month is not None:
    print(f'{current_month}\t{total:.2f}')
> EOF
sky@MS97377:~$ chmod +x reducer_monthly_sales.py
sky@MS97377:~$ hdfs dfs -rm -r /user/$USER/sales/monthly_output
rm: '/user/sky/sales/monthly_output': No such file or directory
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
put /use> -input /user/$USER/sales/input/sales_data.csv \
-output /user/$USER/sales/monthly_output \
-per "py" -mapper "python3 mapper_monthly_sales.py" \
-reducer "python3 reducer_monthly_sales.py" \
-file mapper_monthly_sales.py \
reducer_m> -file reducer_monthly_sales.py
2026-07-25 16:51:25,726 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_monthly_sales.py, reducer_monthly_sales.py, /tmp/hadoop-unjar1272362965461028408/] [] /tmp/streamjob5214398192632652585.jar tmpDir=null
sky@MS97377:~
> EOF
sky@MS97377:~$ chmod +x reducer_monthly_sales.py
sky@MS97377:~$ hdfs dfs -rm -r /user/$USER/sales/monthly_output
rm: '/user/sky/sales/monthly_output': No such file or directory
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
put /use> -input /user/$USER/sales/input/sales_data.csv \
-output /user/$USER/sales/monthly_output \
-per "py" -mapper "python3 mapper_monthly_sales.py" \
-reducer "python3 reducer_monthly_sales.py" \
-file mapper_monthly_sales.py \
reducer_m> -file reducer_monthly_sales.py
2026-07-25 16:51:25,726 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_monthly_sales.py, reducer_monthly_sales.py, /tmp/hadoop-unjar1272362965461028408/] [] /tmp/streamjob5214398192632652585.jar tmpDir=null
2026-07-25 16:51:27,476 INFO client.DefaultHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:51:27,906 INFO client.DefaultHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:51:28,310 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1784977574676_0002
2026-07-25 16:51:29,421 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 16:51:30,014 INFO mapreduce.JobSubmitter: number of splits:2
2026-07-25 16:51:30,380 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1784977574676_0002
2026-07-25 16:51:30,380 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-25 16:51:30,691 INFO conf.Configuration: resource-types.xml not found
2026-07-25 16:51:30,692 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-25 16:51:30,854 INFO impl.YarnClientImpl: Submitted application application_1784977574676_0002
2026-07-25 16:51:30,934 INFO mapreduce.Job: The url to track the job: http://MS97377.localdomain:8088/proxy/application_1784977574676_0002/
2026-07-25 16:51:30,937 INFO mapreduce.Job: Running job: job_1784977574676_0002
2026-07-25 16:51:42,178 INFO mapreduce.Job: Job job_1784977574676_0002 running in uber mode : false
2026-07-25 16:51:42,184 INFO mapreduce.Job: map 0% reduce 0%
2026-07-25 16:51:55,633 INFO mapreduce.Job: map 100% reduce 0%
2026-07-25 16:52:03,714 INFO mapreduce.Job: map 100% reduce 100%
2026-07-25 16:52:05,812 INFO mapreduce.Job: Job job_1784977574676_0002 completed successfully
2026-07-25 16:52:06,016 INFO mapreduce.Job: Counters: 55

File System Counters
  FILE: Number of bytes read=960791
  FILE: Number of bytes written=2759856
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=2339808
  HDFS: Number of bytes written=249
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2

```

```

sky@MS97377: ~
Bytes Written=240
2026-07-25 16:52:06,016 INFO streaming.StreamJob: Output directory: /user/sky/sales/monthly_output
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/monthly_output/part-00000
2024-01 33355727.82
2024-02 31315836.66
2024-03 32336259.82
2024-04 31991068.42
2024-05 32759417.04
2024-06 31404894.48
2024-07 31801249.00
2024-08 32570495.00
2024-09 31599182.57
2024-10 31858390.99
2024-11 30376362.06
2024-12 32671178.38
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/monthly_output/part-00000 | sort -t'$' -k2 -rn | head -1
2024-01 33355727.82
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/monthly_output/part-00000
2024-01 33355727.82
2024-02 31315836.66
2024-03 32336259.82
2024-04 31991068.42
2024-05 32759417.04
2024-06 31404894.48
2024-07 31801249.00
2024-08 32570495.00
2024-09 31599182.57
2024-10 31858390.99
2024-11 30376362.06
2024-12 32671178.38
sky@MS97377:~$ cat > mapper_product_qty.py << 'EOF'
> #!/usr/bin/env python3
>
> import sys
>
> for line in sys.stdin:
>     line = line.strip()
>
>     if not line:
>         continue
>
>     parts = line.split(',')
>
>     if len(parts) != 6:
>
sky@MS97377:~
>         continue
>
>     parts = line.split(',')
>
>     if len(parts) != 6:
>         continue
>
try:>
>     try:
>         product = parts[1]
>         quantity = int(parts[5])
>
>         print(f"{product}\t{quantity}")
>
>     except ValueError:
>         continue
>
> EOF
sky@MS97377:~$ chmod +x mapper_product_qty.py
sky@MS97377:~$ cat > reducer_product_qty.py << 'EOF'
> #!/usr/bin/env python3
>
> import sys
>
> current_product = None
> total_qty = 0
> line in>
> for line in sys.stdin:
> line = >     line = line.strip()
>     if n>
>     if not line:
> contin>         continue
>
>     product, qty = line.split('\t')
>     qty = int(qty)
>
>     if product == current_product:
>         total_qty += qty
>     else:
>         if current_product is not None:
>             print(f"{current_product}\t{total_qty}")
>
>         current_product = product
>         total_qty = qty
>
> if current_product is not None:

```

```

sky@MS97377: ~
> if product == current_product:
>     total_qty += qty
> else:
>     if current_product is not None:
>         print(f"{current_product}\t{total_qty}")
>
>     current_product = product
>     total_qty = qty
>
> if current_product is not None:
>     print(f"{current_product}\t{total_qty}")
F> EOF
sky@MS97377:~$ chmod +x reducer_product_qty.py
sky@MS97377:~$ hdfs dfs -rm -r /user/$USER/sales/product_output
rm: '/user/sky/sales/product_output': No such file or directory
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
> -input /user/$USER/sales/input/sales_data.csv \
> -output /user/$USER/sales/product_output \
> -mapper "python3 mapper_product_qty.py" \
> -reducer "python3 reducer_product_qty.py" \
> -file mapper_product_qty.py \
> -file reducer_product_qty.py
2026-07-25 16:55:53,441 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_product_qty.py, reducer_product_qty.py, /tmp/hadoop-unjar352305982816121374/] [] /tmp/streamjob5840322654972513553.jar
2026-07-25 16:55:55,018 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:55:55,403 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 16:55:55,775 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1784
2026-07-25 16:55:56,455 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 16:55:56,985 INFO mapreduce.JobSubmitter: number of splits:2
2026-07-25 16:55:57,612 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1784977574676_0003
2026-07-25 16:55:57,612 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-25 16:55:58,030 INFO conf.Configuration: resource-types.xml not found
2026-07-25 16:55:58,031 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-25 16:55:58,328 INFO impl.YarnClientImpl: Submitted application application_1784977574676_0003
2026-07-25 16:55:58,641 INFO mapreduce.Job: The url to track the job: http://MS97377.localdomain:8088/proxy/application_1784977574676_0003/
2026-07-25 16:55:58,657 INFO mapreduce.Job: Running job: job_1784977574676_0003
2026-07-25 16:56:07,957 INFO mapreduce.Job: Job job_1784977574676_0003 running in uber mode : false
2026-07-25 16:56:07,960 INFO mapreduce.Job: map 0% reduce 0%
2026-07-25 16:56:22,590 INFO mapreduce.Job: map 50% reduce 0%
2026-07-25 16:56:23,612 INFO mapreduce.Job: map 100% reduce 0%
2026-07-25 16:56:29,698 INFO mapreduce.Job: map 100% reduce 100%
2026-07-25 16:56:32,791 INFO mapreduce.Job: Job job_1784977574676_0003 completed successfully
sky@MS97377:~
2026-07-25 16:56:33,070 INFO streaming.StreamJob: Output directory: /user/sky/sales/product_output
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/product_output/part-00000
Headphones 18363
Keyboard 19021
Laptop 18936
Monitor 18647
Mouse 18771
Printer 18848
Scanner 18633
Webcam 18740
sky@MS97377:~$ hdfs dfs -cat /user/$USER/sales/product_output/part-00000 | sort -t$'\t' -k2 -rn | head -5
Keyboard 19021
Laptop 18936
Printer 18848
Mouse 18771
Webcam 18740
sky@MS97377:~$ hdfs dfs -get /user/$USER/sales/output/part-00000 city_sales_results.txt
sky@MS97377:~$ ls -l city_sales_results.txt
-rw-r--r-- 1 sky sky 200 Jul 25 16:57 city_sales_results.txt
sky@MS97377:~$ cat > visualize_sales.py << 'EOF'
> import pandas as pd
> import matplotlib
> matplotlib.use('Agg')
> import matplotlib.pyplot as plt
>
> # Load city sales results
> df = pd.read_csv(
>     'city_sales_results.txt',
>     sep='\t',
>     header=None,
>     names=['City', 'TotalSales', 'TransactionCount']
> )
>
> print("\n=== Sales by City Summary ===")
> print(df)
>
> # Calculate average transaction value
> df['AvgTransaction'] = df['TotalSales'] / df['TransactionCount']
>
> # Plot top cities
> top_cities = df.nlargest(10, 'TotalSales')
>
> plt.figure(figsize=(12,6))
> plt.bar(top_cities['City'], top_cities['TotalSales']/1_000_000)

```

```
sky@MS97377: ~  
> print(df)  
>  
> # Calculate average transaction value  
> df['AvgTransaction'] = df['TotalSales'] / df['TransactionCount']  
>  
> # Plot top cities  
> top_cities = df.nlargest(10, 'TotalSales')  
>  
> plt.figure(figsize=(12,6))  
> plt.bar(top_cities['City'], top_cities['TotalSales']/1_000_000)  
>  
> # Print top cities by Sales (Mill> plt.title("Top 10 Cities by Sales (Millions ₹)")  
> plt.xlabel("City")  
> plt.ylabel("Sales (Millions ₹)")  
> plt.xticks(rotation=45)  
>  
> plt.tight_layout()  
> plt.savefig("sales_analysis.png")  
>  
> print("\n Visualization saved as sales_analysis.png")  
> EOF  
sky@MS97377:~$ python3 visualize_sales.py  
  
=== Sales by City Summary ===  
   City  TotalSales  TransactionCount  
0  Ahmedabad  48144482.44             6232  
1  Bangalore  48713073.14             6300  
2   Chennai  48259632.40             6231  
3    Delhi  47550288.23             6188  
4  Hyderabad  47306662.01             6243  
5   Kolkata  47815272.70             6256  
6   Mumbai  47276682.25             6174  
7     Pune  48973969.97             6376  
  
 Visualization saved as sales_analysis.png  
sky@MS97377:~$ ls -lh sales_analysis.png  
-rw-rw-r-- 1 sky sky 33K Jul 25 16:58 sales_analysis.png  
sky@MS97377:~$ cmd.exe /c start sales_analysis.png  
cmd.exe: command not found  
sky@MS97377:~$ cp sales_analysis.png /mnt/c/Users/<YourWindowsUsername>/Desktop/  
-bash: YourWindowsUsername: No such file or directory  
sky@MS97377:~$  
  
sky@MS97377:~$ ls -lh sales_analysis.png  
-rw-rw-r-- 1 sky sky 33K Jul 25 16:58 sales_analysis.png  
sky@MS97377:~$ cmd.exe /c echo %USERNAME%  
cmd.exe: command not found  
sky@MS97377:~$ ls /mnt/c/Users  
'All Users'  Default  'Default User'  Public  desktop.ini  sky  
sky@MS97377:~$ cp sales_analysis.png /mnt/c/Users/sky/Desktop/  
sky@MS97377:~$
```

5. Conclusion

This project successfully demonstrates the use of Hadoop Streaming with Python MapReduce for processing large-scale food delivery data. A synthetic dataset containing thousands of food orders was generated and stored in HDFS. Using custom mapper and reducer programs, city-wise revenue, monthly sales trends, and product quantity analysis were performed efficiently. The results were further visualized using Python libraries such as Pandas and Matplotlib to derive meaningful business insights. Overall, the project highlights the practical application of Big Data Analytics, Hadoop Ecosystem, Python Programming, and Data Visualization for analyzing large datasets in a distributed environment.